



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Computer Science and Engineering
Summer Term Examination
Academic Year: 2024-25

Class: SY CSE	Subject: Digital Electronics	Subject Code: ESC282 / PCC-CS203
Date: 20 June 2025	Time: 14:30 to 17:30	Max. Marks: 100

Q.N.	Questions	Max ks	CO	BT
Q 1	a) Convert the decimal number 756.625 to its equivalent in: i) Binary ii) Octal iii) Hexadecimal	6	CO1	BT1
	b) Simplify the following Boolean expression using Karnaugh Map (K-Map) and implement using logic gates: $F(A, B, C, D) = \sum(0, 1, 2, 5, 6, 7, 8, 9, 10, 14)$	8	CO2	BT3
	c) Prove the following Boolean identity: $AB + A'B + AB' = A + B$	6	CO3	BT4
Q 2	a) Design a 3-to-8 line decoder using two 2-to-4 decoders. Show the block diagram and explain. (8 Marks)	8	CO4 CO2	BT5
	b) Design a full adder circuit using two half adders and an OR gate. Derive the expressions and show logic diagram.	6	CO3	BT4
	c) Realize the following function using an 8:1 multiplexer: $F(A, B, C) = \sum(0, 2, 3, 5, 6)$	6	CO4	BT3
Q 3	a) Explain the working of JK flip-flop with logic diagram, truth table, and characteristic equation.	6	CO3 CO4	BT1
	b) Design a 3-bit synchronous up counter using T flip-flops. Show logic diagram and timing diagram.	8	CO1	BT2
	c) Differentiate between asynchronous and synchronous counters. Give one example of each.	6	CO2	BT3
Q 4	a) Design a 4-bit shift register and explain its working for right-shift operation.	6	CO1 CO4	BT4
	b) Explain the following types of memories with suitable diagrams: i) SRAM ii) DRAM iii) ROM iv) EEPROM	8	CO3	BT1
	c) What is the difference between serial-in serial-out and parallel-in parallel-out shift registers? Explain with diagrams.	6	CO4	BT3
Q 5	a) Define and explain the basic logic gates (AND, OR, NOT) with truth tables and circuit symbols.	6	CO1	BT1
	b) Explain the concept of universal gates. Realize AND, OR, and NOT gates using only NAND gates.	8	CO1	BT1
	c) Draw the logic circuit and truth table for the expression: $Y = (A + B') \cdot (C + A')$	6	CO3	BT3

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Explain history of evolution of microprocessors, architecture of 8086, 8088, 8087
CO 2	Write 8086 assembly language programs using the 8086 instruction set.
CO 3	Write modular programs using procedures and macros and write stack and interrupt programming.
CO 4	Interface 8086 to static memory chips and 8255, 8254, 0808 ADC, 0800 DAC, Keyboard, display and stepper motors.
CO 5	Use INT 21 DOS interrupt function call to handle keyboard and display.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Computer Science
Summer Term Examination
Academic Year: 2024-25

Class: SY (CSE)

Course: Discrete Mathematics

Course Code: CS201

Date: 23/06/2025

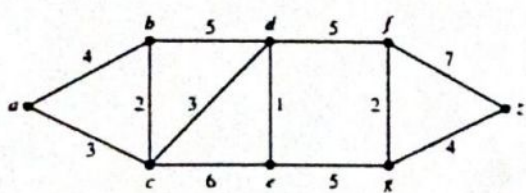
Time: 02:30 to 05:30

Max. Marks: 100

Note:

- Assume suitable data whenever necessary and state it clearly.
- Attempt **ALL** questions from the following.
- This paper contains 3 pages.

Q.N.		Marks	CO	BT
Q.1	Give the answer of the following questions. (Each Carries 2 Marks)	10		
a	$p \rightarrow q$ is logically equivalent to _____ a) $\neg p \vee \neg q$ b) $p \vee \neg q$ c) $\neg p \vee q$ d) $\neg p \wedge q$	2	CO1	BT1
b	What is the Cardinality of the Power set of the set $\{0, 1, 2\}$? a) 8 b) 6 c) 7 d) 9	2	CO2	BT2
c	Every Isomorphic graph must have _____ representation. a) cyclic b) adjacency list c) tree d) adjacency matrix	2	CO5	BT1
d	_____ is not necessarily a property of a Group. a) Commutativity b) Existence of inverse for every element c) Existence of Identity d) Associativity	2	CO3	BT1
e	In the principle of mathematical induction, which of the following steps is mandatory? a) induction hypothesis b) inductive reference	2	CO2	BT2

		c) induction set assumption d) minimal set representation			
Q.2		Use Warshall's algorithm to find the transitive closures of the relations on $A = \{a, b, c, d, e\}$. $R = \{(b, c), (b, e), (c, e), (d, a), (e, b), (e, c)\}$	10	CO2	BT3
Q.3		Define Partial order relation with example. What is the Hasse diagram or Poset? Draw the Hasse diagram for divisibility on the set – $\{1, 2, 3, 6, 12, 24, 36, 48\}$.	10	CO2	BT1, BT3
Q.4		Thirteen people on a softball team show up for a game. a) How many ways are there to choose 10 players to take the field? b) How many ways are there to assign the 10 positions by selecting players from the 13 people who show up? c) Of the 13 people who show up, three are women. How many ways are there to choose 10 players to take the field if at least one of these players must be a woman?	10	CO3	BT3, BT4
Q.5	a	Construct a Binary Search Tree by inserting integers in order 50, 15, 62, 5, 20, 58, 91, 3, 8, 37, 60, 24. Also find preorder , inorder , postorder traversal of Binary Search Tree.	10	CO5	BT6
Q.6	a	Show that $A = \{0, 3, 6, 9, 12\}$ with binary operation $(A, +15)$ is an abelian group or not. Also check whether it is a cyclic group or not, find all generators.	10	CO3	BT2, BT3
Q.7	a	Find the shortest path between a and z in the given weighted graph (using Dijkstra's algorithm) 	10	CO5	BT3
Q.8		a) Show that the premises "If you send me an e-mail message, then I will finish writing the program," "If you do not send me an e-mail message, then I will go to sleep early," and "If I go to sleep early, then I will wake up feeling refreshed" lead to the conclusion "If I do not finish writing the program, then I will wake up feeling refreshed." b) Show that the premises "It is not sunny this afternoon and it is colder than yesterday," "We will go swimming only if it is sunny," "If we do not go swimming, then we will take a canoe trip," and "If we	10	CO1	BT3

		take a canoe trip, then we will be home by sunset" lead to the conclusion "We will be home by sunset."			
Q.9		Difference Between Graph and Tree. (With example)	10	CO5	BT1,BT2
Q.10		Prove that if n is an integer and $3n + 2$ is even, then n is even using a) a proof by contraposition. b) a proof by contradiction.	10	CO2	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	For a given logic sentence express it in terms of predicates, quantifiers, and logical Connectives.
CO 2	For a given a problem, derive the solution using deductive logic and prove the solution based on logical inference.
CO 3	For a given mathematical problem, classify its algebraic structure.
CO 4	Evaluate Boolean functions and simplify expressions using the properties of Boolean algebra.
CO 5	Develop the given problem as graph networks and solve with techniques of graph theory.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Computer Science and Engineering Department
Summer Term Examination
Academic Year: 2024-25

Class: SY CSE

Course: Data structure

Course Code: PCC-CS206

Date: 25.06.2025

Time: 14:30 to 17:30

Max. Marks: 100

Note: i. Solve in sequence only ii. Figures to right indicate full marks.

Q.N.		Marks	CO	BT
Q.1	Solve the following [2 Marks each]	20		
a	Define data structure. Mention any two applications of data structures?		CO3	BT1,2
b	Explain different types of data structures.		CO2	BT2
c	Write the postfix and prefix notations for the following expression: $A/B * C - D * E + F/G$		CO3	BT5
d	Define Queue? What are the types of Queues?		CO1,2	BT6
e	List the steps in pre order traversal.		CO1,4	BT6
f	State the properties of a Binary Tree?			
g	If 2, 7, 6, 10, 9, 8, 15, 17, 20, 19, 16, 12 is the postorder traversals of a binary search tree. Then this tree's preorder traversal is:		CO1,4	BT3,6
h	What is searching?		CO1,4	BT1
i	What is the best case and worst case time complexity of bubble sort and insertion sort?		CO1	BT4,5
j	Level of a node is distance from root to that node. For example, level of root is 1 and levels of left and right children of root is 2. The maximum number of nodes on level i of a binary tree is In the following answers, the operator '^' indicates power.		CO5	BT6
	A. $2^{(i-1)}$ B. 2^i C. $2^{(i+1)}$ D. $2^{[(i+1)/2]}$			
Q.2	Solve the Following [5 Marks each]	20		
a	Define BFS with an example.		CO3	BT3
b	What are the various operations performed on the Queue? Write algorithm for any ONE.		CO3,5	BT4
c	Define tree. List and explain tree terminologies.		CO4	BT5
d	What is the difference between the arrays and linked list? Give any two applications of each.		CO3,5	BT4
Q.3	Explain the following operations in a doubly linked list: (a) Create an empty list. (b) Insert the elements 10 and 20 at the front of the list. (c) Insert the elements 30 at the middle of the list. (d) Insert the elements 15, 45 at the end of the list. (e) Delete the middle element from the list.	10	CO3	BT2,6
Q.4	Write a program to perform basic operations on stack.	10	CO1,2	BT6
Q.5	Construct a binary search tree from the given values. Consider the first value as the root value. Values: 45, 23, 29, 85, 92, 7, 11, 35, 49, 51	10	CO4	BT5
Q.6	What is an AVL Tree? How does it differ from a Binary tree?	10	CO4	BT1

Q.7	What is collision? List various collision resolution techniques. Explain any two collision resolution techniques.	10	CO4	BT2
Q.8	What is meant by sorting? Write an algorithm for insertion sort and illustrate with an example.	10	CO1,4	BT1

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO1: Understand the concept of Dynamic memory management, data types, algorithms and its analysis.

CO2: Understand basic data structures and system specific use.

CO3: Understand non-linear data structures and their design and use.

CO4: Apply Algorithm and data structures for solving problems.

CO5: Design and implementation of new data structures.